

YUDONG LUO

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EDUCATION

Carnegie Mellon University

Dec 2026

Master of Science in Artificial Intelligence Engineering - Mechanical Engineering

Concentration: **Robotic and Control Systems**

Selected Coursework: Robot Learning; Computer Vision; AI Engineering Tool Chains; Machine Learning; Deep Learning

University of Electronic Science and Technology of China (UESTC)

Jun 2025

Bachelor of Engineering in Communications Engineering

SKILLS

Languages: Python, C/C++, MATLAB

Robotics & Simulation: ROS, Isaac Sim/Lab, MuJoCo, PCL, Open3D, OpenCV

ML & Tools: PyTorch, CUDA, Docker, Linux, Git

PROJECTS & RESEARCH EXPERIENCE

EgoX: Egocentric Cross-Embodiment Learning for Versatile Manipulation | CMU × Bosch Research

Apr 2026 – Present

Safe AI Lab | Co-author, manuscript in preparation (target: ICRA 2027)

- Contributed to a cross-embodiment framework that pretrains a shared Transformer trunk on Apple Vision Pro–captured human egocentric interactions and teleoperated demonstrations from heterogeneous robots, then finetunes embodiment-specific policies on target-robot data.
- Developed the R1 Lite embodiment integration and deployment: implemented the robot controller and data-collection pipeline, and co-designed how its camera, proprioceptive, end-effector, and action modalities are tokenized and routed through the shared MXT trunk.
- Designed real-robot tasks, collected demonstrations, and built rollout and evaluation workflows; on xArm, cross-embodiment pretraining raised Scoop success from 61.5% to 84.6% (task score 72.3% → 96.9%) and OOD Sort from 1/7 to 5/7.

Skill-Specialized Diffusion Policy with Task-Space Residual RL | CMU

May 2026 – Present

IROS 2026 Robotic Assembly Competition, RoCo Task Board | Team of 3

- Designed a skill-specialized diffusion policy for 9-part sequential assembly: a shared visual-state encoder feeds three DDPM action heads grouped by end-phase motion, FiLM-conditioned on part type, pick-place coordinates, and release mode.
- Formulated a task-space residual RL layer for mm-tolerance snap parts: a lightweight policy outputs a clipped 6-DoF Cartesian correction mapped to joint space via a damped Jacobian pseudo-inverse, pretrained on demonstration residuals and finetuned with TD3 in Isaac Sim, reaching 3.79 mm / 2.95° insertion accuracy.
- Benchmarked monolithic baselines on the same suite — full-parameter finetuned $\pi 0.5$ at 2/9, a single diffusion policy at 1/9, ACT at 0/9 — showing one policy cannot cover both open-release and snap-fit parts, motivating the skill-specialized design.
- Separately benchmarked Flow Matching against Diffusion Policy on Push-T; Beta-distribution timestep sampling raised single-step-inference success from 0.13 to 0.82, cutting generation cost for real-time control.

Semantic-Aware Robotic Grasping and VLA Fine-tuning from Human Egocentric Videos | CMU

Jan 2026 – Jun 2026

Personal Project

- Built an egocentric hand-object interaction pipeline: 3D reconstruction with MoGe2, hand pose in MANO parameter space with HoWoR, plus contact-point and object-pose estimation; auto-labeled segmentation and grasps with Qwen3 VLM+SAM3.
- Designed a decoupled human-to-robot mapping: approach pose and contact regions from MANO hand poses serve as semantic priors, while grasp configurations are generated by Contact-GraspNet in gripper space, bridging the human-hand / parallel-jaw morphology gap.
- Built a three-layer grasping pipeline — Perception (RealSense D405 → Moondream2 VLM → EdgeTAM) → Planning (Contact-GraspNet) → Execution (IK → trajectory) — reaching 83% success across 5 object types, up to ~20% gain with human priors.

Distilling a 32B Vision-Language Model to 2B for Edge Deployment | CMU

Jan 2026 – Jun 2026

- Built a 50k-sample teacher-data pipeline with vLLM and ablated four distillation objectives against two parameter budgets using DeepSpeed ZeRO-2 and PEFT, targeting spatial reasoning within a Jetson-class compute budget.
- Located the binding constraint at 2B scale as catastrophic forgetting rather than teacher signal strength: full fine-tuning fell 29 points below the untuned student, while LoRA $r=64$ reached 75.40% on CV-Bench.

Learning-based Multi-Camera Visual Odometry | CMU

Sep 2025 – Mar 2026

Research Assistant, AirLab × Denso

- Generalized MAC-VO to arbitrary N-camera rigs via configurable camera-pair matching and cross-view triangulation, unifying pretrained optical flow networks (VGGT, UFM, RoMaV2) into one uncertainty-aware frontend.
- Implemented segmentation-based motion masking to filter dynamic objects and used the masks to train a keypoint selector, improving VO robustness in dynamic scenes.

PROFESSIONAL EXPERIENCE

HiGaAs Microwave Technology Co., Ltd.

Jul 2024 – Aug 2024

Testing and Automation Intern, System-level Packaging Product Department

Chengdu, China

- Built an automated instrument-control and data-acquisition system for chip reliability testing, driving benchtop equipment over SCPI/Socket and automating temperature cycling, logging, and analysis.

PATENT

Qin, L.; Luo, Y. *Nighttime Traffic Scene Salient Object Detection with Improved YOLOv9*. CN App. 2024119132237, 2024. Patent Pending.